



Modelling Financial Markets

Chapter 19: Review and Project Presentations



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CV: Cross-Validation
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EVT: Extreme Value Theory
- ▣ FDR: False Discovery Rate
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GBM: Geometric Brownian Motion
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ IGARCH: Integrated GARCH
- ▣ IQR: InterQuartile Range
- ▣ LRMEs: Long-Run Marginal Expected Shortfall
- ▣ MDE: Minimum Detectable Effect
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SML: Security Market Line
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ VOLARE: VOLatility Archive for Realized Estimates (database)

Chapter Route and Outcomes

Route of this chapter

- Part I: the course in one picture, a review of Chapters 0–18, six pitfalls, and one integrated case study, from raw BET prices to a backtested value-at-risk (VaR) at 1%
- Part II: the team project, AI in the project, a research case study on nonstandard errors, one pass of the AI discovery loop
- Technical detail sits in the Appendix at the end

After this chapter you can

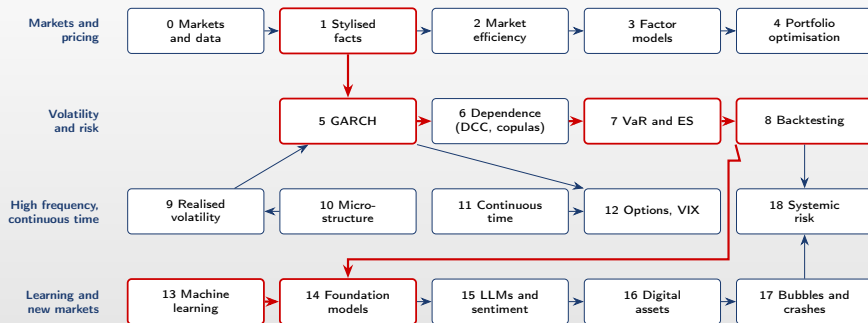
- run one question from data to a backtested risk forecast with valid inference
- recognise the six pitfalls in your own code
- write a project plan with replication, identification, power and a specification grid



Bucharest University of Economic Studies (ASE),
Piața Romană

Photo: [Joe Mabel \(2014\); CC BY 3.0; Wikimedia Commons](#)

The Course in One Picture



- Red: the main thread, from stylised facts (Chapter 1: heavy tails, volatility clustering, almost no linear predictability) to risk forecasts we can test (Chapters 5–8: GARCH gives σ_{t+1} , VaR and ES the loss, backtesting says whether to trust it)
- Chapters 9–12 measure volatility better (intraday data, option prices) and in continuous time; Chapters 13–15 bring flexible models, always judged with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system; special Chapter 20: realised volatility with path signatures on the VOLARE data

Five Questions for Every Model

Before estimating

- ▣ **Data:** which series, which calendar, which adjustments (dividends, splits, holidays)?
- ▣ **Model:** which stylised fact does it capture, and which does it ignore?
- ▣ **Estimation:** on which window, with which information at time t ?

After estimating

- ▣ **Evaluation:** out of sample, with a loss that matches the target (QLIKE, FZ0) and the right diagnostic tests (e.g. Kupiec)?
- ▣ **Uncertainty:** robust standard errors, bootstrap, a test against a simple benchmark?
- ▣ If one answer is missing, the result is not yet a finding

Chapters 0–2: Markets, Returns and Efficiency

- ▣ **Chapter 0, Financial Markets in 2026:** Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
 - ▶ 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year with 17.6% volatility; Bitcoin 60.6% a year with 66.7%
 - ▶ BET, July 2007 to February 2009: a drawdown of -82.5%
- ▣ **Chapter 1, Stylised Facts and Returns:** Returns are uncorrelated but not independent: their magnitude is predictable, but their sign is largely unpredictable
 - ▶ BET, Hill index of losses (top $k = 153$): $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03]: the fourth moment is likely infinite
 - ▶ Days beyond 5σ are more than 1000 times more frequent than the Normal distribution predicts
- ▣ **Chapter 2, Market Efficiency:** Efficiency: prices fully reflect the information set; a test of abnormal-return predictability is a joint test of efficiency and the expected-return model [Fama \(1970\)](#)
 - ▶ Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
 - ▶ Statistical rejection is not exploitable profit

Chapters 3–5: Asset Pricing, Portfolios and GARCH

- ▣ **Chapter 3, Factor Models and Asset Pricing:** Expected returns should reward only exposure to systematic factors
 - ▶ 25 size and book-to-market portfolios: SML slope -4.4% , $GRS = 4.18$: the CAPM is rejected
 - ▶ Sector ETFs: the first principal component (the market) explains 66% of the variance
- ▣ **Chapter 4, Portfolio Optimisation:** Markowitz asks the right question, but estimation error dominates the answer [Markowitz \(1952\)](#)
 - ▶ 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
 - ▶ 96% of the estimated tangency portfolios are truly worse than $1/N$
- ▣ **Chapter 5, Conditional Volatility: GARCH:** Volatility clusters, so today's shock predicts tomorrow's variance [Engle \(1982\)](#), [Bollerslev \(1986\)](#)
 - ▶ S&P 500, 2000–2026: $\alpha + \beta = 0.9947$, a half-life of 131 days
 - ▶ BET: profile-likelihood 95% CI for $\alpha + \beta$ is $[0.975, 1]$ (IGARCH not rejected): the half-life is at least 27 days, with no finite upper bound

Chapters 6–8: Dependence, Risk Measures and Backtesting

- **Chapter 6, Multivariate Volatility and Dependence:** Correlations move with the macro regime; diversification is not guaranteed
 - ▶ SPY–TLT (long US Treasury bond ETF): ρ from -0.40 to 0.11 after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
 - ▶ Bitcoin–SPY DCC correlation: 0.04 before 2020, 0.31 since
- **Chapter 7, Value-at-Risk and Expected Shortfall:** VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
 - ▶ S&P 500, daily: historical $\text{VaR}_{1\%} = 3.45\%$, $\text{ES}_{1\%} = 5.01\%$
 - ▶ The Normal distribution gives only 2.80%: heavy tails matter
- **Chapter 8, Backtesting and Evaluating Risk Forecasts:** Test the number and the timing of breaches: coverage and independence [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
 - ▶ S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS 1.67%, FHS 1.23%, GARCH-EVT 1.13% breaches
 - ▶ A GARCH filter plus a flexible tail wins in all four markets

Chapters 9–11: High Frequency, Liquidity and Continuous Time

- ▣ **Chapter 9, Realised Volatility:** Intraday prices measure daily variance (RV , realised variance); the log-HAR (heterogeneous autoregressive) model captures its long memory [Corsi \(2009\)](#)
 - ▶ SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by $\sqrt{RV_t}$
 - ▶ log-HAR beats GARCH(1,1)- t in QLIKE by 10% for SPY ($p = 0.021$) and 22% for Bitcoin
- ▣ **Chapter 10, Market Microstructure:** Prices form in limit order books; the spread pays for processing, inventory and adverse selection
 - ▶ Amihud ratio (bp per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
 - ▶ SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar
- ▣ **Chapter 11, Continuous-Time Models:** Brownian motion is the limit of random walks; Itô's lemma solves the price equations
 - ▶ S&P 500 against geometric Brownian motion: 62 days beyond ± 4 standard deviations, against 0.59 expected
 - ▶ Hill index of S&P 500 losses: 2.87 [2.07, 3.67] (Chapter 16): the fourth moment is likely infinite, unlike the Normal log returns of GBM

Chapters 12–14: Options, Machine Learning and Foundation Models

- ▣ **Chapter 12, Options and the Volatility Surface:** Black–Scholes prices options by hedging; implied volatility is the market's view of σ [Black & Scholes \(1973\)](#)
 - ▶ 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
 - ▶ The variance risk premium is positive on 86% of days, but not in crashes
- ▣ **Chapter 13, Machine Learning in Finance:** Flexible models, but little signal and much noise; the finance-specific part is the process [López de Prado \(2018\)](#)
 - ▶ Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
 - ▶ S&P 500: the fractionally differenced series is stationary at $d^* = 0.25$; no model beats “always up”
- ▣ **Chapter 14, Deep Learning and Time-Series Foundation Models:** LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
 - ▶ SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH- t 0.283; the first two are not significantly different
 - ▶ Raw Chronos-2 VaR 1% on the S&P 500: 1.67% breaches

Chapters 15–16: Text and Digital Assets

- ▣ **Chapter 15, LLMs and Sentiment Analysis:** From dictionaries to FinBERT and large language models (LLMs): text becomes a number
 - ▶ Qwen 7B vs 1.5B on financial headlines: +21.5 percentage points in accuracy, McNemar $p < 0.001$
 - ▶ A headline-sentiment strategy: Sharpe +0.63 before and −1.48 after costs; break-even cost 1.5 bp per trade
- ▣ **Chapter 16, Digital Assets and DeFi:** Cryptocurrencies form a distinct asset class in the feature space of [Pele et al. \(2023\)](#), where a tail factor separates them; their Hill tail indices are not significantly lower than those of equities or gold (Chapter 16)
 - ▶ Bitcoin–S&P 500 link: sup-Wald 20.2 (5% critical value 11.6), break estimated on 9 Mar 2020, not at the 2024 ETF launch
 - ▶ Hill index of losses: Bitcoin 2.73 [2.12, 3.33], S&P 500 2.87 [2.07, 3.67]: statistically indistinguishable

Chapters 17–18: Bubbles and Systemic Risk

- **Chapter 17, Bubbles and Crashes:** A bubble is a price path that grows faster than fundamentals can justify
 - ▶ Bitcoin 2014–2026 (weekly): GSADF 3.85 against a 95% critical value of 2.27: explosive episodes are detected
 - ▶ One vs two Markov regimes for Bitcoin: $LR = 123.2$ against a parametric-bootstrap 95% critical value of 11.8 ($p = 0.005$)
- **Chapter 18, Systemic Risk and Networks:** Systemic risk: losses of one institution that spread to the whole system
 - ▶ ΔCoVaR 1% of eleven banks: 4 of 25 within-region pairs differ at 5%, 0 after Holm; overlap of intervals alone would have ranked none
 - ▶ Simulated LRMES (22 days, market below -10%): JPMorgan 10.6% today, 23.7% with the information of 31 March 2020

Pitfalls of Design: Look-Ahead Bias and Overfitting

- **Look-ahead bias:** any parameter, threshold or scaling estimated on the full sample leaks the future
 - ▶ Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set; the case study measures the effect on BET
 - ▶ Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
 - ▶ Remedy: rolling or expanding windows; purging and embargo; write the information set $\mathcal{F}_{t-1}$ next to every forecast
- **Overfitting:** the in-sample fit rewards noise that will not repeat
 - ▶ Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
 - ▶ Backtest overfitting: trying many strategies and keeping the best [Bailey et al. \(2017\)](#); the deflated Sharpe ratio corrects for the number of trials and for non-normal returns [Bailey & López de Prado \(2014\)](#)
 - ▶ Remedy: fix the design before looking at the test period; report every model you tried; compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall of Design: Multiple Testing

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$, and the best reaches $|t| \approx 3.06$
 - ▶ This is why the factor literature asks for $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery
- Remedy: count your tests and correct
 - ▶ Bonferroni and Holm control the chance of any false discovery (FWER); Benjamini–Hochberg controls their expected share (FDR)
 - ▶ For forecasts: the model confidence set instead of many pairwise tests [Hansen, Lunde & Nason \(2011\)](#)

Pitfall of Data: Joining on Returns Instead of Prices

- ❑ Bitcoin trades every day, SPY only on weekdays
 - ▶ Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
 - ▶ Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- ❑ Bitcoin, 2020–2026, on the common days: cumulative return 1025% (right) against 680% (wrong)
 - ▶ Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong join hides weekend risk
- ❑ Remedy: align prices on common days first; one series alone stays on its own calendar

Pitfall of Inference: Heavy Tails Ignored

- Classical standard errors assume independence and finite fourth moments
 - ▶ BET losses: Hill tail index $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03] [Hill \(1975\)](#): with $\alpha < 4$ the population kurtosis is infinite
 - ▶ So the sample kurtosis (11.1) estimates nothing, and a bootstrap interval for it is not valid [Athreya \(1987\)](#): report the tail index instead
- BET GARCH persistence: the Wald CI [0.974, 1.007] crosses the stationarity boundary $\alpha + \beta = 1$, where the Normal approximation fails [Andrews \(1999\)](#)
 - ▶ Profile-likelihood CI inside the parameter space $p \leq 1$: [0.975, 1] (IGARCH, $p = 1$, not rejected against the boundary critical value 2.71), so the half-life is at least 27 days and has no finite upper bound
- Remedy: robust (HAC) errors, block bootstrap where it is valid, tail indices instead of kurtosis, boundary-aware intervals

Pitfall of Evaluation: a Loss That Cannot Rank the Target

- Variance forecasts are compared with a noisy proxy (r_t^2 or RV_t)
 - ▶ With a conditionally unbiased proxy, MSE and QLIKE rank forecasts as the true variance would; QLIKE depends only on the ratio proxy/forecast, so volatile days weigh less than under MSE [Patton \(2011\)](#)
 - ▶ Chapter 9: log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive
- ES alone has no strictly consistent scoring function on the usual broad classes of distributions; the pair (VaR, ES) has one, under mild conditions [Fissler & Ziegel \(2016\)](#)
 - ▶ Use FZ0 with Diebold–Mariano tests [Diebold & Mariano \(1995\)](#); expect finite-sample limits on precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Remedy: choose the loss before the comparison, and state it in the project report

One Question, Four Chapters

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6 670 daily log returns
- Duplicate records (rows identical to the previous one) and weekend-dated rows removed; trading days with an unchanged close kept

The five steps

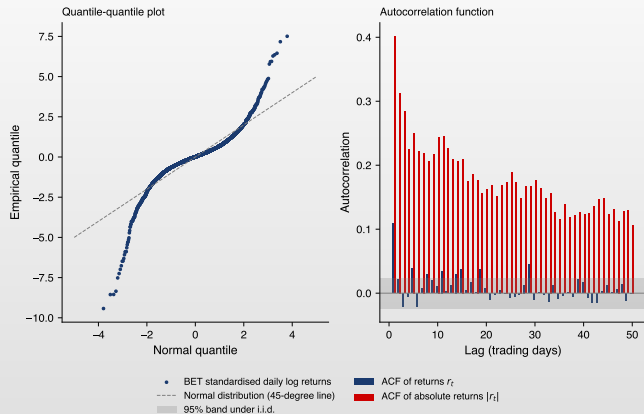
- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student- t innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones
- Formal comparison of the four models: FZ0 loss, DM tests, model confidence set



The Stock Exchange Palace, Bucharest (2023)

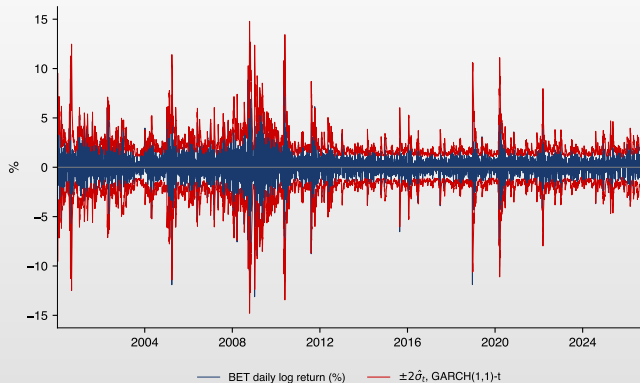
Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

Step 1: Stylised Facts of the BET



- Heavy tails (left, quantiles against the Normal distribution): Hill index of losses $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03], $k = 153$; below 4, so the kurtosis is not reported as an estimate
- 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution; worst day -13.1% (log) on 7 Jan 2009
- Clustering (right): first-order autocorrelation of $|r_t|$ 0.40, slow decay; Ljung–Box $Q(10)$ on r_t^2 2440; returns themselves 0.11: a less liquid market (Chapters 2, 10)
- Consequence: a Normal, constant-volatility VaR will be too low in stress and too high in calm periods

Step 2: GARCH(1,1)- t Volatility



- BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student- t innovations, 2000–2026; robust standard errors: $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$: heavy tails even after filtering
- Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$, a half-life of about 74 trading days; profile-likelihood 95% CI [0.975, 1]: persistence cannot be bounded away from 1 (pitfall of inference)
- Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample 1.67%

Step 3: Four VaR 1% Forecasts

Unconditional (last 500 days)

- ▣ **HS** (historical simulation): the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
- ▣ **Normal**: $\text{VaR} = -(\hat{\mu} + \hat{\sigma} z_{0.01})$, $z_{0.01} = -2.326$

Conditional (GARCH- t on the last 1000 days)

- ▣ **GARCH- t** :

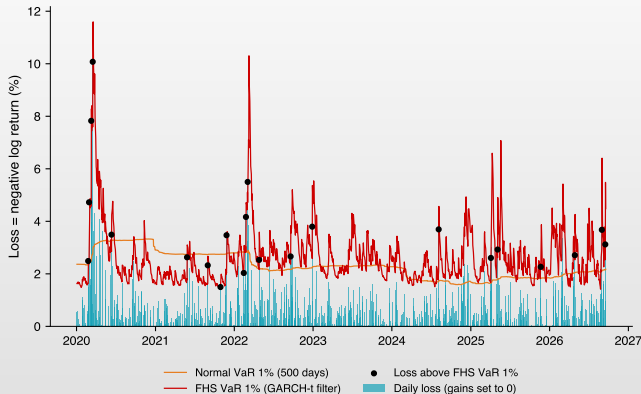
$$\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t t_\nu^{-1}(0.01) \sqrt{(\nu - 2)/\nu})$$
- ▣ **FHS** (filtered historical simulation):

$$\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \hat{q}_{0.01}(z))$$
, the 1% quantile of the standardised residuals [McNeil & Frey \(2000\)](#)

Real-time design

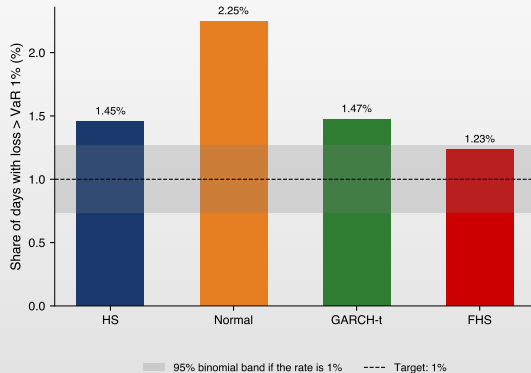
- ▣ Forecast for day t uses data up to $t - 1$ only
- ▣ GARCH re-estimated every 21 trading days; $\hat{\sigma}_t$ updated daily
- ▣ Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
- ▣ Expected breaches at 1%: about 54

Step 3: Losses and VaR 1%, 2020–2026



- Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR
- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash; the Normal VaR on a 500-day window moves only slowly
- 1 675 days, about 17 breaches expected: FHS 22, Normal 41; Normal breaches cluster in the first weeks of every stress episode
- Last forecast: FHS 4.48%, Normal 2.18%: the models disagree most when volatility changes

Step 4: Breach Rates



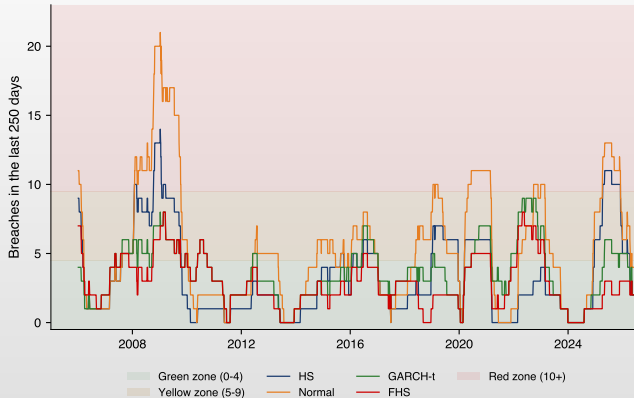
- Share of days with a loss above the VaR 1% forecast, 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$
- Grey band: the 95% binomial range [0.74%, 1.27%] if the true rate is 1%
- Only FHS (1.23%) lies inside the band; the Normal model (2.25%) breaches more than twice as often as it should
- Next: are the differences significant, and are the breaches independent?

Step 4: Coverage and Independence Tests

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

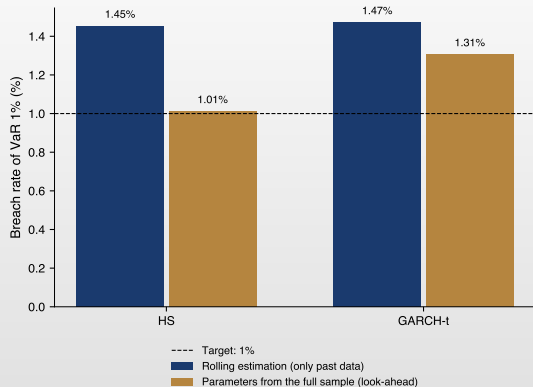
- p_{uc} : Kupiec coverage test; p_{ind} : Christoffersen independence test; p_{cc} : both jointly [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- Only FHS is rejected by none of the three tests at 5%; HS and Normal also fail independence: their breaches cluster
- GARCH- t : independence is not rejected, but there are too many breaches: the Student- t tail is not heavy enough at 1%

Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- The Normal model spends 18.6% of the windows in the red zone, HS 5.5%, the two GARCH-filtered models never

The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction
- GARCH- t with full-sample parameters: 1.31% of days, with rolling estimation 1.47%
- Both look better than their real-time versions: the gap is the value of information nobody had
- Only the rolling result is evidence about a forecasting model

Step 5: Three Tools for a Formal Comparison

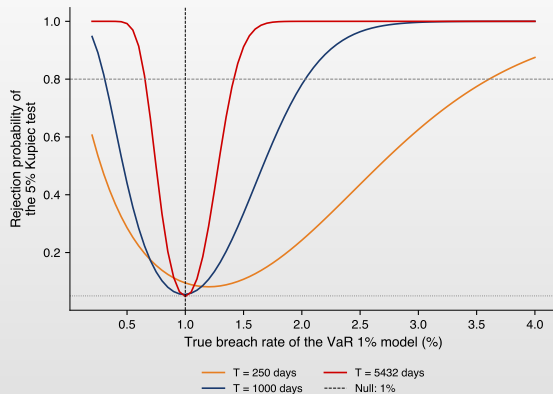
- **Dynamic Quantile (DQ)** test: regress $I_t - 0.01$ on a constant, four lagged hits and VaR_t ; $DQ \sim \chi_6^2$ Engle & Manganelli (2004)
 - ▶ It detects breaches that are predictable from the past or from the level of VaR, which the Christoffersen test (one lag) can miss
- **FZ0 score** for the pair (VaR 2.5%, ES 2.5%), $v = -\text{VaR}$, $e = -\text{ES}$, return y :
$$S = -\frac{1\{y \leq v\}(v-y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$$
 Patton, Ziegel & Chen (2019)
 - ▶ Consistent for (VaR, ES) Fissler & Ziegel (2016); pairwise Diebold–Mariano tests with HAC variance Diebold & Mariano (1995); the model confidence set at 10% Hansen, Lunde & Nason (2011)
- **ES backtest**: comparative, against HS as the standard model; red if the model is significantly worse, green if significantly better, yellow otherwise Nolde & Ziegel (2017)
 - ▶ All forecasts use only data up to $t - 1$; 5 432 days, 3 Jan 2005 – 18 Sep 2026

Step 5: Results

Model	DQ	p_{DQ}	FZ0	p_{MCS}	ES zone
HS	192.1	<0.001	1.424	0.00	standard
Normal	364.6	<0.001	1.495	0.00	red
GARCH-t	17.9	0.007	1.175	0.18	green
FHS	9.9	0.129	1.157	1.00	green

- Only FHS passes the DQ test at 5% ($p = 0.129$); GARCH- t fails it ($p = 0.007$) through its excess of breaches
- FZ0 (lower is better): FHS and GARCH- t are not significantly different (DM $t = 1.39$), both beat HS and Normal ($|t| > 4.9$); the 10% model confidence set is {GARCH- t , FHS}
- Comparative ES backtest: Normal red, GARCH- t and FHS green: the ranking of Step 4 survives formal inference

How Much Can a Backtest Detect? The Power of Kupiec



- Exact binomial probability that the 5% Kupiec test rejects a VaR 1% model, by its true breach rate
- $T = 250$ (one Basel year): a true rate of 2% is detected only 24% of the time; rejection needs 0 or at least 7 breaches
- Minimum detectable rate (MDE) with 80% power: 3.65% at $T = 250$, 2.05% at $T = 1000$, 1.45% at $T = 5432$; a true 1.3% is detected with probability 55% only
- A non-rejection with low power is “no evidence”, not “no problem”: state the power before the test

The Project in Four Steps

- ▣ An empirical study of one question about financial markets, answered with the methods of this course; **70%** of the final grade (quizzes and seminar activity 20%, attendance 10%, no final exam)
- ▣ **1. Replicate:** reproduce one full table or figure of a published paper (or of the lecture), cell by cell
 - ▶ Report the discrepancy of every cell and explain it; an AI assistant cannot guess a published table
- ▣ **2. Extend:** new data, a new market (BVB, crypto) or a more recent sample, under a **pre-analysis plan** committed before the test period is opened
 - ▶ The plan states the null, the test, the loss, the sample split, the identification strategy, the power of the main test and the specification grid with its correction
- ▣ **3. Deliver on GitHub:** code, lockfile, fixed seeds, data provenance; one notebook regenerates every number
 - ▶ [AI_USE.md](#) and [AI_ERRORS.md](#) are mandatory; the report is an 8–12-page paper
- ▣ **4. Present and defend:** team presentation, then a 10-minute oral defence
 - ▶ Each member explains one result and answers a “what changes if...” question; grades can differ between members

Replication Standard

- Target: one full table or figure, with its DOI, page and table number, not a single coefficient
 - ▶ Report a cell-by-cell comparison: published value, your value, difference, and the reason for each gap (sample, filter, data vendor, definition)
- Large-scale replications show why: many anomalies shrink out of sample or with better data [Hou, Xue & Zhang \(2020\)](#), while most factors replicate once the design is right [Jensen, Kelly & Pedersen \(2023\)](#)
- Reproducibility: a lockfile of package versions, fixed random seeds, series names and dates of the data, one command that regenerates every number
 - ▶ Different teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#): more reproducible work is associated with a smaller gap (research case study, this chapter)

Choosing the Question

A good question is

- ▣ **Specific:** one market, one period, one quantity
- ▣ **Testable:** a null hypothesis and a benchmark
- ▣ **Feasible:** data you can obtain and clean in a few weeks (course data in data/market; public sources: BNR, FRED, Kenneth French Data Library, DefiLlama)
- ▣ **Relevant:** an answer matters to an investor, a bank or a regulator

Examples

- ▣ Too vague: "Is Bitcoin risky?"
 - ▶ Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin, and which model forecasts it best since 2024?"
- ▣ Too broad: "Can machine learning predict the market?"
 - ▶ Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability	Chapters 1–2: moments, robust tests, variance ratios
Expected returns, factors, portfolios	Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR
Risk measures and capital	Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with ML, deep learning or text	Chapters 13–15: purged CV, foundation models, sentiment
Crypto, bubbles, systemic risk	Chapters 16–18: stablecoins, GSADF, CoVaR, networks

- Starting points: the Part C question of every seminar (table in the Appendix); your own question is welcome if it passes the five questions of this chapter

Identification: from Correlation to a Claim

- Event questions (“did spot ETFs, the FTSE upgrade or a bank tax change X?”) need a counterfactual
 - ▶ A control market not hit by the event: difference-in-differences on the risk measure
 - ▶ A synthetic control built from several untreated markets [Abadie, Diamond & Hainmueller \(2010\)](#); placebo event dates (e.g. $\pm$ one year)
- Unknown timing: a sup-Wald break test [Andrews \(1993\)](#); report the estimated break date, not an imposed one
- Forecasting questions: out of sample only, with a test period declared in the plan
- One event is $n = 1$: say what the design can and cannot identify

Power and Robustness without p-Hacking

- **Power:** before the main test, compute the smallest effect the design detects with 80% power at 5% (MDE)
 - ▶ Deliverable: a simulated power curve of the main test at your T ; short crypto, ETF or post-event samples make many questions underpowered (Kupiec at $T = 250$: MDE 3.65%)
- The pre-analysis plan is a timestamped commit in the repository, made before the test period is opened
- Specification curve: estimate every defensible combination of window, lags, α and sample start; plot all estimates sorted, with their intervals [Simonsohn, Simmons & Nelson \(2020\)](#)
 - ▶ Report the spread across specifications, not the best cell
- Correct the family: Holm for any false discovery, Benjamini–Hochberg for their share, or a bootstrap reality check
 - ▶ False and missed discoveries in finance [Harvey & Liu \(2020\)](#); the scientific outlook of the field [Harvey \(2017\)](#)

Forecast Evaluation for Risk Projects

- ▣ Volatility: compare forecasts with a noisy, conditionally unbiased proxy only through a robust loss (MSE or QLIKE) [Patton \(2011\)](#)
 - ▶ Many models: the model confidence set [Hansen, Lunde & Nason \(2011\)](#), not the lowest average loss
- ▣ VaR: coverage, independence and the DQ test [Engle & Manganelli \(2004\)](#); with a benchmark, conditional predictive ability tests [Giacomini & White \(2006\)](#)
 - ▶ State the power of every backtest at your T
- ▣ ES: score the pair (VaR, ES) with FZ0; comparative backtests with three zones [Nolde & Ziegel \(2017\)](#); dynamic (VaR, ES) models [Patton, Ziegel & Chen \(2019\)](#)
 - ▶ The integrated case study (Steps 1–5) is the template

The Report and the Presentation

The report: an 8–12-page paper

- ▣ **1. Introduction:** question, hypothesis, contribution;
- 2. Literature:** 10–15 clickable references, at least 3 from the last five years, including [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#)
- ▣ **3. Data and replication:** sources, period, cleaning rules; the replicated table cell by cell
- ▣ **4. Design:** plan, identification, power, benchmark, loss, tests
- ▣ **5. Results** with uncertainty; **6. Robustness:** the specification curve with the correction; **7. Conclusion:** the answer in one sentence, and its limits

The presentation (about 12 minutes)

- ▣ Question (1 slide); data and one chart that shows the problem (2); method and benchmark (2); main result with its uncertainty (2–3); robustness, limits, answer (2)
- ▣ One message per slide; readable axes, legend below the chart
- ▣ Every member presents and can answer questions on the code; then the 10-minute oral defence, without AI

Project Grading and Timeline

Component	What we look for
Replication (step 1)	A published table or figure reproduced cell by cell; every gap explained
Question, data, extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Plan, identification, power, real-time design, benchmark, specification curve with a correction
Interpretation and report	Economic meaning, limitations, honest conclusion; 8–12 pages, 10–15 references
GitHub repository (step 3)	One command reproduces every result (lockfile, seeds); AI_USE.md, AI_ERRORS.md
Presentation and defence (step 4)	Individual: one result explained, answers to “what changes if...”

- Week 1 after this chapter: team, question, one-page proposal (Seminar 19, C1), GitHub repository; weeks 2–3: data, replication (step 1), plan committed
- Weeks 4–5: extension, evaluation, robustness (step 2); final weeks: repository with README and the two AI files (step 3), presentation and oral defence (step 4); exact dates are announced at the course and seminar

The AI Policy and the Two Mandatory Files

Policy

- ▣ **Allowed:** AI assistants for code, debugging, literature search and writing
- ▣ **Declared:** undeclared use of AI counts as plagiarism
- ▣ **Checked:** you are responsible for every line; a reference that does not exist counts as fabricated data
- ▣ **No AI during the oral defence:** “the AI wrote it” is not an explanation

The two files

- ▣ **AI_USE.md:** tools and versions, who used them for what, where AI-written code or text was kept, how you verified the replicated table, data, formulas and references
- ▣ **AI_ERRORS.md:** at least **three** places where the assistant was wrong: what you asked, what it answered, what was wrong, how you found out, the correct version
- ▣ Errors you caught are a strength of the project, not a weakness

Typical AI Errors in Financial Modelling

- ▣ **Look-ahead:** a rolling statistic or a scaler that includes day t in the forecast for day t
 - ▶ Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- ▣ **Sign of VaR:** the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
 - ▶ Check: $\text{VaR}_\alpha = -q_\alpha(r)$ is positive whenever $q_\alpha(r) < 0$; a one-day VaR 1% of an index is a few percent
- ▣ **Annualisation:** volatility multiplied by 252 instead of $\sqrt{252}$, or 252 days used for crypto that trades every day
 - ▶ Check: means scale with h , volatilities with $\sqrt{h}$; use the real number of trading days (365 for crypto)
- ▣ **Random cross-validation (CV) on time series:** shuffled K-fold trains on the future (69.4% accuracy on pure noise, Chapter 13)
 - ▶ Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo [López de Prado \(2018\)](#)
- ▣ **Invented references,** wrong degrees of freedom, library defaults you did not choose, numbers stated without running code
 - ▶ Check: search every DOI on [Crossref](#) and find the claim in the paper; every seminar has a C2 exercise that trains this

How Much Do Researchers' Choices Matter?

- ▣ Menkveld et al. (2024), *Journal of Finance* 79(3): 164 research teams test the same six hypotheses on the same data (Finance Crowd Analysis Project, #fincap; Sec. I.A)
- ▣ **Standard error**: uncertainty from sampling, the data-generating process; **nonstandard error**: uncertainty from the analysis path each researcher picks, the evidence-generating process (Introduction)
 - ▶ Model: team j reports $X_j = e_j + \varepsilon_j$, e_j the researcher fixed effect of its path, ε_j the sampling error (Sec. I.C, eq. 1)
 - ▶ Nonstandard error $:= Q_{0.75}(x) - Q_{0.25}(x)$, the interquartile range (IQR) of the estimates across teams (eq. 3); the IQR, not the standard deviation, because the e_j can have fat tails
- ▣ **Data**: all trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records; six hypotheses of “no annual trend” (market efficiency, realised spread, client share of volume, ...); each team reports the average yearly change in % with its standard error (SE)
 - ▶ Four stages in 2021: independent analysis, peer feedback, the five best papers shared, final estimates; design details in the Appendix
- ▣ Link to this chapter: every team project is one path through the “garden of forking paths”; pre-registration, the specification grid and a repository that reruns every number are the remedies

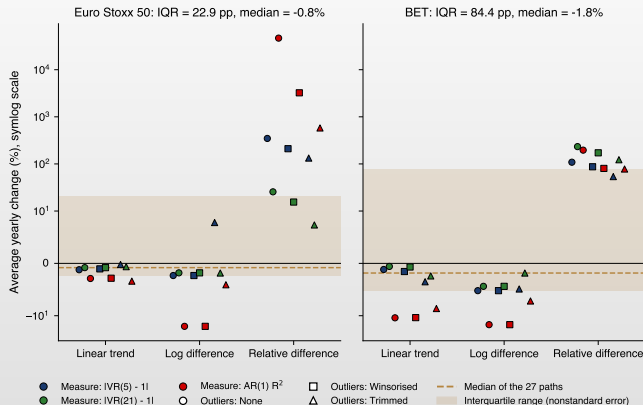
Key Results: Size and Quality

- ▣ **Size:** for market efficiency the median estimate is -1.1% per year, with an IQR of 6.7 percentage points (pp); even for the client share of volume, a simple ratio, the IQR is 1.2 pp around -3.3% (Table I, Panel C)
 - ▶ The median SE of 2.5% implies a Gaussian IQR of $1.35 \times 2.5 = 3.4\%$, half the 6.7 pp; the no-dispersion null is rejected for all six hypotheses, $p < 0.0001$ (Table II)
- ▣ **Quality:** a rise of one standard deviation in reproducibility is associated with a 25.0% lower IQR, the same rise in peer rating with a 33.3% lower IQR (Table III, Fig. 2)
 - ▶ Code that reruns end to end is associated with answers closer to the consensus

Key Results: Feedback, Beliefs and Forks

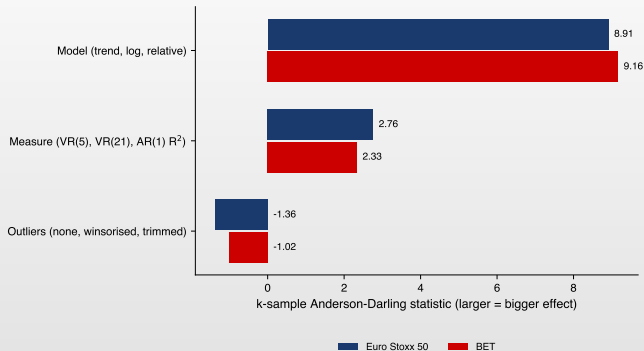
- ▣ **Feedback:** from stage 1 to stage 4 the IQR falls by 47.2% and the interdecile range by 68.2% (Table IV, Fig. 3)
 - ▶ Peer review narrows the dispersion but does not remove it
- ▣ **Beliefs:** pooled over hypotheses, the teams' predicted standard deviation of estimates is 71.7% below the realised one (Sec. II.B.3, Fig. 4)
 - ▶ Researchers underestimate how much teams analysing the same data can differ
- ▣ **Forks:** model choice and sampling frequency move the estimates most; a relative-change model gives a median annual trend of $-23\,000\%$ on daily data, -200% monthly and -4.56% yearly (Fig. 6, Fig. 7)
 - ▶ Jensen's inequality for relatives $M_t = X_t/X_{t-1}$: $\mathbb{E} \ln M_t \leq \ln \mathbb{E} M_t$; log differences add up exactly across frequencies and a linear trend is linear in the data, so neither has this frequency wedge

A Mini-Multiverse on Course Data: 27 Paths per Index



- Daily closes 2002–2018 (data/market); outliers $\times$ measure $\times$ model forks of Table V; the band is the IQR, the nonstandard error of the multiverse
- IQR: 22.9 pp (Euro Stoxx 50) and 84.4 pp (BET), around medians of -0.8% and -1.8% a year; $|VR - 1|$ and R^2 measure departures from a random walk, so a positive trend means declining efficiency
- The model fork creates the dispersion: over the nine linear-trend paths the IQR is only 2.0 pp and 7.5 pp; relative differences explode (medians $+210.4\%$ and $+108.7\%$, up to 46 823%), as in the paper
- Against $1.35\times$ the median SE (25.9 and 50.8 pp), the IQR exceeds the sampling benchmark for the BET, not for the Euro Stoxx 50

Which Fork Moves the Estimates?



- *k*-sample Anderson–Darling statistic of the 27 estimates grouped by the alternative at each fork (Sec. II.C): model 8.91 (Euro Stoxx 50) and 9.16 (BET); measure 2.76 and 2.33; outliers -1.36 and -1.02
- Same ranking as the paper (Fig. 6): model choice moves the estimates most, on a large euro-area index and on a frontier index
- The 27 estimates share the same data, so the statistic is a descriptive ranking, not a test; a negative value is not evidence of no effect
- For your project: fix the model of the yearly change in the plan (log differences or linear trends), then report the whole grid and the fork that drives it

Limits and an Open Extension

- ▣ **Limits stated by the authors:** the multiverse covers a few forks only and explains the reported estimates poorly (footnote 32); beliefs were asked as standard deviations, not IQRs (footnote 28)
 - ▶ The test of dispersion assumes consistent reported SEs (Sec. I.C); the dynamics of journal refereeing are kept out of scope (Sec. I.A)
 - ▶ A multiverse spans only the options in its grid; its IQR neither estimates nor bounds from below the dispersion across real teams
- ▣ **Data limitation of the replication:** daily closes only, so the intraday frequency pairs of Table V cannot be run (Seminar 19, C3 extends it to the last full year)
- ▣ **Open extension:** does a pre-registered grid shrink the nonstandard error of a risk question? Measure the IQR of VaR 1% conclusions for the BET across defensible paths
 - ▶ For your project: report the distribution over your grid and the forks that move it most, not a single cell

An Open Question and the Discovery Loop

- ▣ **Open question:** how much of a result is the data, and how much the researcher? Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries
Jensen, Kelly & Pedersen (2023)
 - ▶ AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
 - ▶ Formal version: the conclusion must hold over a pre-registered set $\mathcal{S}$ of defensible specifications, with the family corrected (FWER or FDR); report its distribution over $\mathcal{S}$, not the best cell
- ▣ **Loop:** question and H_0/H_1 (week 1); literature and the replication target seen in the PDF (weeks 1–2); pre-registration committed before the first result (week 2); replication as a passing test in the repository (weeks 2–3); the whole grid (weeks 4–5); an AI examiner before the defence
 - ▶ ‘Here are all the cells we ran. Which conclusions survive a Holm correction, and which would a referee call data snooping?’
- ▣ **Checks by the researcher:** the target number is in the paper; the plan precedes the first results file in the commit history; every run is logged and counted; a fresh clone regenerates every number; each member can derive the key formula without AI
 - ▶ Prompts for every step and the test for each check: Appendix

Mini-Case: Replicate, Then Extend

- **Replicate:** FHS VaR 1% for the BET; Chapter 8 reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
 - ▶ The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$); same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal, reading each p -value alone at 5%: “only GARCH-EVT is correctly calibrated on all four assets”
- **Extend:** the 24 conditional-coverage tests of Chapter 8 (4 assets $\times$ 6 models) form one family
 - ▶ Rejections at 5%: 12 without correction, 10 with Holm (FWER), 11 with BH (FDR 5%)
 - ▶ FHS fails only on the S&P 500 ($p = 0.040$); within the family of 24 it survives, so after Holm FHS and GARCH-EVT pass on all four assets
- Interpretation: “FHS fails for the S&P 500” depends on whether the test is read alone or in its family; pre-register the family and report both

Your Turn: the Capstone Plan

- ▣ **Question:** your seed from Part C, written as H_0/H_1 , with the observation that would falsify it
- ▣ **Replicate first:** one published table seen in the paper, turned into a passing test in the repository
 - ▶ Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- ▣ **Deliverables** along the timeline: proposal and replication target, pre-registration commit, passing replication test, grid of results, mock defence, report, oral defence
- ▣ **AI log:** the prompts that changed the design in `AI_USE.md`; every caught error, with the test that caught it, in `AI_ERRORS.md` ([templates](#))

Key Takeaways

- ▣ Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- ▣ Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- ▣ Heavy tails change inference: use robust errors and the bootstrap, report intervals, state the power
- ▣ Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- ▣ The analysis path is itself a source of uncertainty: pre-register, replicate, report the whole grid
- ▣ New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Check Yourself and Further Study

Retrieval questions

- Why is the sample kurtosis of BET returns not an estimate of anything?
- Why can the Wald interval for GARCH persistence not be reported, and what replaces it?
- What does a non-rejection of the Kupiec test at $T = 250$ tell you?
- Why is the IQR of a 27-path multiverse not the nonstandard error of 164 teams?

Further study

- Textbooks: Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)
- Advanced courses on quantinar.com:  Statistics of Financial Markets;  Measuring Statistical Risk
- Code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_19

Appendix: Key Inference by Chapter (1/7)

- **Chapter 0:** Data quality is a modelling decision: source, adjustment and calendar change the result
 - ▶ For i.i.d. log returns, annualised by the number of periods a year: $\text{se}(\hat{\mu}_{\text{ann}}) = \sigma_{\text{ann}}/\sqrt{Y}$ over Y years, whatever the frequency; $\text{se}(\hat{\sigma}) \approx \sigma\sqrt{(K-1)/(4T)}$ over T days, K the population kurtosis (finite fourth moment needed); serial dependence calls for long-run variances
 - ▶ SPY 2015–2026: $\text{se}(\hat{\mu}_{\text{ann}}) \approx 5.1\%$ for a mean of 13.7%: means are barely estimable, volatilities are
- **Chapter 1:** Heavy tails and volatility clustering in every market [Cont \(2001\)](#)
 - ▶ Ljung–Box $Q(m) \sim \chi_m^2$ needs i.i.d. returns; under volatility clustering it over-rejects: use the robust Q^* with summands $T\hat{\rho}_k^2/\hat{\tau}_k$ [Lobato, Nankervis & Savin \(2001\)](#)
 - ▶ Tails: Hill $\hat{\alpha}$ with $\text{se} \approx \hat{\alpha}/\sqrt{k}$ [Hill \(1975\)](#); if $\alpha < 4$ the sample kurtosis is not consistent and its bootstrap fails [Athreya \(1987\)](#)
- **Chapter 2:** Efficiency varies over time (adaptive markets); tests need robust inference
 - ▶ Heteroskedasticity-robust variance ratio $z^*(q) \Rightarrow N(0, 1)$ under a martingale difference null [Lo & MacKinlay \(1988\)](#)
 - ▶ Several horizons q at once: the joint test $\max_q |z^*(q)|$ with studentised maximum modulus critical values [Chow & Denning \(1993\)](#); rolling windows need a multiplicity correction too

Appendix: Key Inference by Chapter (2/7)

□ **Chapter 3:** Hundreds of published factors require higher hurdles [Harvey, Liu & Zhu \(2016\)](#)

- ▶ $GRS = \frac{T-N-K}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{\mathbf{1} + \hat{\mu}_f^\top \hat{\Omega}^{-1} \mathbf{1} \hat{\mu}_f} \sim F(N, T - N - K)$ only under i.i.d. Normal errors [Gibbons, Ross & Shanken \(1989\)](#); N assets, K factors; $\hat{\Sigma}$, $\hat{\Omega}$: residual and factor covariances, divisor T
- ▶ Fama–MacBeth s.e. ignore that betas are estimated: with $c = \hat{\gamma}^\top \hat{\Sigma}_f^{-1} \hat{\gamma}$, Shanken's covariance is $(1 + c)(\hat{V}_{FM} - \hat{\Sigma}_f/T) + \hat{\Sigma}_f/T$ for estimated factor loadings [Shanken \(1992\)](#)

□ **Chapter 4:** Shrinkage, constraints and risk-based rules stabilise the weights; $1/N$ is a hard benchmark [DeMiguel, Garlappi & Uppal \(2009\)](#)

- ▶ $se(\widehat{SR}) \approx \sqrt{(1 + \widehat{SR}^2/2)/T}$ for i.i.d. Normal returns, per period [Lo \(2002\)](#); serial correlation changes the annualisation factor
- ▶ Is portfolio A better than $1/N$? Test $SR_A - SR_B = 0$ with HAC or studentised bootstrap inference [Ledoit & Wolf \(2008\)](#), not by comparing two point estimates

□ **Chapter 5:** Persistence close to 1; asymmetry in equities; heavy-tailed innovations

- ▶ QMLE: $\sqrt{T}(\hat{\theta} - \theta) \Rightarrow N(0, A^{-1}BA^{-1})$ (sandwich); Hessian-only s.e. are wrong for non-Normal innovations [Bollerslev & Wooldridge \(1992\)](#)
- ▶ Near $\alpha + \beta = 1$ the parameter is close to the boundary of the covariance-stationary region: Wald intervals are unreliable; use profile likelihood or bootstrap restricted to the parameter space [Andrews \(1999\)](#)

Appendix: Key Inference by Chapter (3/7)

- **Chapter 6:** Copulas separate the margins from dependence; tail dependence drives joint losses
 - ▶ DCC is estimated in two steps: second-step s.e. must account for the first-step volatilities [Engle \(2002\)](#)
 - ▶ Crisis vs calm correlation: ρ rises mechanically with volatility; test on $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, δ the relative rise in variance [Forbes & Rigobon \(2002\)](#)
- **Chapter 7:** Filtered historical simulation and conditional EVT react to volatility in days [McNeil & Frey \(2000\)](#)
 - ▶ Historical VaR, i.i.d. returns with a continuous density $f(q_\alpha) > 0$: $\sqrt{T}(\hat{q}_\alpha - q_\alpha) \Rightarrow N(0, \alpha(1 - \alpha)/f(q_\alpha)^2)$; in the tail f is small, so the estimate is noisy; under dependence $\alpha(1 - \alpha)$ becomes the long-run variance of the hits $\mathbb{1}\{r_t \leq q_\alpha\}$
 - ▶ ES averages the αT largest losses: with αT of a few dozen and a tail index near 3, its variance is dominated by one or two days
- **Chapter 8:** Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests [Fissler & Ziegel \(2016\)](#)
 - ▶ Kupiec $LR_{uc} \sim \chi_1^2$, Christoffersen $LR_{ind} \sim \chi_1^2$; the DQ test regresses $I_t - \alpha$ on lagged hits and VaR_t , $DQ \sim \chi_k^2$ [Engle & Manganelli \(2004\)](#)
 - ▶ Power is low: at $T = 250$ the 5% Kupiec test detects a true rate of 2% only 24% of the time; compare models with conditional predictive ability tests [Giacomini & White \(2006\)](#)

Appendix: Key Inference by Chapter (4/7)

- **Chapter 9:** Returns divided by realised volatility have a distribution close to the Normal distribution
 - ▶ HAR errors are heteroskedastic and, for multi-day targets that overlap, serially correlated: use HAC s.e.; forecasts are compared with DM tests on a robust loss [Patton \(2011\)](#)
 - ▶ Microstructure noise biases RV upwards at high frequency: two-scale estimators remove the bias [Zhang, Mykland & Aït-Sahalia \(2005\)](#)
- **Chapter 10:** Liquidity follows the clock and the VIX, and disappears in stress
 - ▶ The Roll spread needs $\text{Cov}(\Delta p_t, \Delta p_{t-1}) < 0$; when the sample covariance is positive the estimator is undefined, so report how often
 - ▶ An illiquidity premium is a cross-sectional claim: Fama–MacBeth with Newey–West s.e. and the Shanken correction [Shanken \(1992\)](#)
- **Chapter 11:** Jumps explain the tails, stochastic volatility the clustering
 - ▶ Normality of GBM log returns: Jarque–Bera $\sim \chi^2_2$ rejects everywhere; jumps are tested with bipower variation, $(RV - BV)$ standardised $\Rightarrow N(0, 1)$ [Barndorff-Nielsen & Shephard \(2006\)](#)
 - ▶ The test is daily: over many days, control the number of false jump days

Appendix: Key Inference by Chapter (5/7)

- **Chapter 12:** Smile and skew encode the fear of crashes
 - ▶ VRP_t uses the next 21 days of realised variance: consecutive values overlap, so the mean needs Newey–West s.e. with at least 21 lags
 - ▶ Implied volatility is a risk-neutral forecast: the premium, not a forecast error, explains most of the gap
- **Chapter 13:** Purged cross-validation, meta-labelling and deflated Sharpe ratios [Bailey & López de Prado \(2014\)](#)
 - ▶ Selection among N strategies: the deflated Sharpe ratio tests $SR > E[\max_N \widehat{SR}]$ under the null of no skill [Bailey & López de Prado \(2014\)](#)
 - ▶ Cross-validation on time series must purge overlapping labels and embargo the test fold, or its accuracy is biased upwards
- **Chapter 14:** Evaluate them with Chapter 8 tools, after the release of their weights
 - ▶ DM test on QLIKE differences with HAC variance: $t_{DM} \Rightarrow N(0, 1)$ for non-nested comparisons [Diebold & Mariano \(1995\)](#)
 - ▶ With many candidate models, the model confidence set controls the family-wise error [Hansen, Lunde & Nason \(2011\)](#)

Appendix: Key Inference by Chapter (6/7)

- **Chapter 15:** Text embeddings enter forecasting models as features
 - ▶ Predictive regression $r_{t+1} = a + bS_t + \varepsilon_{t+1}$: a persistent S_t whose innovations correlate with returns biases $\hat{b}$ and the t -test [Stambaugh \(1999\)](#)
 - ▶ Classifier accuracies on the same texts are paired: compare them with the exact McNemar test, not two separate intervals
- **Chapter 16:** Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance
 - ▶ Crypto factor models: GRS and Fama–MacBeth with the Shanken correction [Gibbons, Ross & Shanken \(1989\)](#), [Shanken \(1992\)](#); heavy tails call for HAC or bootstrap versions
 - ▶ Tails: Hill indices with intervals, not sample kurtosis (Chapter 1); a break in a correlation is tested with an unknown date [Andrews \(1993\)](#)

Appendix: Key Inference by Chapter (7/7)

- **Chapter 17:** Right-tailed unit-root tests date the start and the end of explosive episodes
 - ▶ Under the unit-root null, SADF and GSADF converge to functionals of Brownian motion; critical values depend only on the minimum window r_0 [Phillips, Shi & Yu \(2015\)](#)
 - ▶ Volatility shifts distort the size: use wild-bootstrap critical values; control the multiplicity of date-stamping
- **Chapter 18:** CoVaR (the VaR of the system conditional on the distress of one institution), SRISK and spillover networks measure the contribution of each institution
 - ▶ ΔCoVaR differences between banks: paired tests on joint block-bootstrap draws with a Holm correction; at $\alpha = 1\%$ only about 40 tail days, so extremal-quantile issues
 - ▶ SRISK: LRMEs simulated from GJR-GARCH + DCC [Brownlees & Engle \(2017\)](#); connectedness tables need bootstrap intervals and shrinkage VARs

Appendix: Project Seeds: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	An optimal portfolio of BVB blue chips in RON
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	Capital for a BVB portfolio vs an S&P 500 portfolio

- Each seminar gives the data, a starting method and a reference analysis discussed in class

Appendix: Project Seeds: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?
15	Does headline sentiment pay after costs?
16	Is Bitcoin an alternative asset, a risk asset or a hedge?
17	Is there an AI bubble?
18	How exposed are Romanian banks to euro-area banking stress?

- Chapter 20 (realised volatility with signatures, VOLARE data) also suits a forecasting project

Appendix: Turning a Seed into a Project

- Replicate first: one published table or figure from a paper behind the seed (step 1)
- Narrow it: one market, one period, one number to forecast or test
 - ▶ Example: Seminar 8 becomes “Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 BET–EUR portfolio in 2015–2026?”
- Extend it: a second market, a new model, a longer or more recent sample
 - ▶ Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

Appendix: Data Conventions for the Project

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
 - ▶ Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
 - ▶ Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
 - ▶ Remove holiday fills and weekend quotes where the market is closed; annualise with the real frequency
 - ▶ Several assets: join on prices, then compute returns
- Look for bad prints before modelling: jumps of $\pm 50\%$ in adjusted prices are usually data errors

Appendix: Honest Evaluation Checklist

Before you look at the results

- Write the hypothesis, the benchmark and the loss function
- Fix the estimation window and the test period
- Decide how many models and parameters you will try, and the correction for them
- Write the identification strategy and the power of the main test

When you report

- Out-of-sample results only for claims about forecasting
- Every headline number with a standard error, CI or p -value
- Costs for trading strategies; subperiods and a second market for robustness
- What did **not** work, and why

Appendix: Where Projects Usually Lose Points

- ▣ A question that changes after the results are known
- ▣ In-sample fit presented as forecasting power
- ▣ No benchmark, or a weak one chosen to be beaten
- ▣ Numbers without uncertainty; many tests without correction
- ▣ Charts without units, legends inside the data, unexplained acronyms
- ▣ Code that does not run, or numbers in the report that the code does not produce
- ▣ AI output kept without checks: undeclared use, unchecked code, references that do not exist

Appendix: Nonstandard Errors, Data and Method

- **Data:** all trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records, each side flagged as client or own account (Sec. I.A)
 - ▶ Six hypotheses, each “no annual trend” in: market efficiency, realised bid–ask spread, client share of volume, realised spread on client orders, share of market orders among client orders, gross trading revenue of clients
 - ▶ Each team chooses its own measure and reports the average yearly change in % with its standard error (SE)
- **Four stages** in 2021: independent analysis (paper and code); feedback from two peer evaluators; the five best papers shared; final estimates without the code constraint (Sec. I.A)
 - ▶ A pre-analysis plan was filed publicly before the experiment started (Sec. I.A)
- **Quality:** team quality (first principal component of five proxies), reproducibility score 0–100 by the certification agency Cascad, demeaned peer rating (Sec. I.B)
 - ▶ Quantile regressions of the estimates at the 10th, 25th, 50th, 75th and 90th percentiles; a shrinking IQR means less dispersion (Sec. I.C)
- **Inference:** each estimate is tested against the median estimate, with a Bonferroni correction at a family level of 0.5% (Sec. I.C)
 - ▶ Multiverse analysis: the teams report their choice at each fork of Table V; the k -sample Anderson–Darling statistic ranks the forks by how much they move the estimates (Sec. II.C)

Appendix: Nonstandard Errors, Replication Details

- **Forks from Table V:** outliers (none; the annual measure winsorised or trimmed at its 2.5 and 97.5 percentiles), measure (variance ratio, R^2 of an AR model), frequencies (day to week, day to month), model (linear trend, log difference, relative difference)
 - ▶ Daily data allow the day-to-week and day-to-month comparisons ($VR(5)$, $VR(21)$); the intraday measure frequencies of Table V need intraday data
- **Why relative differences explode:** $m_t = |VR - 1|$ or R^2 is close to zero in some years; the paper finds the same explosion, a median of $-23\,000\%$ on daily data (Fig. 7)
 - ▶ No trend, stationary m_t : the linear slope and the mean log change are centred at zero, but for independent m_t , $\mathbb{E}(m_t/m_{t-1}) = \mathbb{E}m \cdot \mathbb{E}(1/m) \geq 1$ (Jensen), so the mean relative change is positive
- **Anderson–Darling statistic:** the standardised statistic is centred at zero for equal distributions of independent samples; the 27 estimates are strongly correlated, so tabulated critical values and p -values do not apply
 - ▶ Paper, 164 teams: 6.7 pp against 3.4 pp (Table I, Panel C); in the mini-multiverse the IQR exceeds this benchmark for the BET, not for the Euro Stoxx 50

Appendix: The Discovery Loop, Prompts (1/2)

- **Week 1, the question:** sharpen a seed into H_0/H_1 with a replication target
 - ▶ “Here is our seed from Part C of Seminar N. Give three sharper, testable versions, the published table each would replicate and the series in data/market it needs.”
- **Weeks 1–2, literature and target:** the number to replicate is one you have seen in the PDF
 - ▶ “List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample.”
 - ▶ Resolve every DOI on [Crossref](#); search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves
- **Week 2, hypotheses and pre-registration,** committed before the first result
 - ▶ “For our H_0 and H_1 , list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction.”
 - ▶ The commit history of the repository is the evidence that the plan came first

Appendix: The Discovery Loop, Prompts (2/2)

- **Weeks 2–3, code and replication:** the replicated table becomes a test in the repository
 - ▶ “Write a function that reproduces every cell of the published table from data/market and an assertion with the tolerance we accept.”
- **Weeks 4–5, extension and robustness:** run the whole pre-registered grid
 - ▶ “Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?”
- **Before the defence, critique:** the assistant asks, the team answers
 - ▶ “Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them.”
 - ▶ The team answers without AI, as at the defence itself
- **Report:** the distribution of results over the grid, not the best cell; prompts in AI_USE.md, caught errors in AI_ERRORS.md

Appendix: What the Researcher Must Check

- **The replication target exists:** number, table and sample taken from the paper, not from the assistant's summary
 - ▶ Test: the report cites the page and the table; a second member finds the same number there
- **The plan came first**
 - ▶ Test: in the commit history, the pre-registration precedes the first results file
- **Forking paths:** more specifications tried than reported
 - ▶ Test: every run writes its configuration and result to a log; the report states their number and corrects the family
- **Reproducibility:** numbers entered manually in the report
 - ▶ Test: a fresh clone and one command regenerate every number to the reported digit
- **Understanding:** each member can derive the key formula and defend every choice
 - ▶ Test: an internal mock defence, without AI, one week before the real one

References (1/5)

- ▣ Abadie, A., Diamond, A., Hainmueller, J. (2010). [Synthetic control methods for comparative case studies: estimating the effect of California's tobacco control program](#). *Journal of the American Statistical Association*, 105(490), 493–505.
- ▣ Andrews, D.W.K. (1993). [Tests for parameter instability and structural change with unknown change point](#). *Econometrica*, 61(4), 821–856.
- ▣ Andrews, D.W.K. (1999). [Estimation when a parameter is on a boundary](#). *Econometrica*, 67(6), 1341–1383.
- ▣ Athreya, K.B. (1987). [Bootstrap of the mean in the infinite variance case](#). *Annals of Statistics*, 15(2), 724–731.
- ▣ Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). [The probability of backtest overfitting](#). *Journal of Computational Finance*, 20(4), 39–69.
- ▣ Bailey, D.H., López de Prado, M. (2014). [The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality](#). *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2006). [Econometrics of testing for jumps in financial economics using bipower variation](#). *Journal of Financial Econometrics*, 4(1), 1–30.
- ▣ Black, F., Scholes, M. (1973). [The pricing of options and corporate liabilities](#). *Journal of Political Economy*, 81(3), 637–654.
- ▣ Bollerslev, T. (1986). [Generalized autoregressive conditional heteroskedasticity](#). *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). [Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances](#). *Econometric Reviews*, 11(2), 143–172.
- ▣ Brownlees, C., Engle, R.F. (2017). [SRISK: a conditional capital shortfall measure of systemic risk](#). *Review of Financial Studies*, 30(1), 48–79.
- ▣ Chow, K.V., Denning, K.C. (1993). [A simple multiple variance ratio test](#). *Journal of Econometrics*, 58(3), 385–401.

References (2/5)

- ▣ Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the $1/N$ portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R. (2002). Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.

References (3/5)

- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Harvey, C.R. (2017). Presidential address: the scientific outlook in financial economics. *Journal of Finance*, 72(4), 1399–1440.
- ▣ Harvey, C.R., Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5), 2503–2553.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- ▣ Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.

References (4/5)

- ▣ Lo, A.W. (2002). [The statistics of Sharpe ratios](#). *Financial Analysts Journal*, 58(4), 36–52.
- ▣ Lo, A.W., MacKinlay, A.C. (1988). [Stock market prices do not follow random walks: evidence from a simple specification test](#). *Review of Financial Studies*, 1(1), 41–66.
- ▣ Lobato, I., Nankervis, J.C., Savin, N.E. (2001). [Testing for autocorrelation using a modified Box–Pierce Q test](#). *International Economic Review*, 42(1), 187–205.
- ▣ López de Prado, M. (2018). [Advances in Financial Machine Learning](#). Wiley.
- ▣ Markowitz, H. (1952). [Portfolio selection](#). *Journal of Finance*, 7(1), 77–91.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). [Quantitative Risk Management: Concepts, Techniques and Tools](#), revised ed. Princeton University Press.
- ▣ Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). [Nonstandard errors](#). *Journal of Finance*, 79(3), 2339–2390.
- ▣ Nolde, N., Ziegel, J.F. (2017). [Elicitability and backtesting: perspectives for banking regulation](#). *Annals of Applied Statistics*, 11(4), 1833–1874.
- ▣ Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). [Information entropy and measures of market risk](#). *Entropy*, 19(5), 226.

References (5/5)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Phillips, P.C.B., Shi, S., Yu, J. (2015). [Testing for multiple bubbles: historical episodes of exuberance and collapse in the S&P 500](#). *International Economic Review*, 56(4), 1043–1078.
- ▣ Shanken, J. (1992). [On the estimation of beta-pricing models](#). *Review of Financial Studies*, 5(1), 1–33.
- ▣ Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). [Specification curve analysis](#). *Nature Human Behaviour*, 4(11), 1208–1214.
- ▣ Stambaugh, R.F. (1999). [Predictive regressions](#). *Journal of Financial Economics*, 54(3), 375–421.
- ▣ Tsay, R.S. (2010). [Analysis of Financial Time Series](#), 3rd ed. Wiley.
- ▣ Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). [A tale of two time scales: determining integrated volatility with noisy high-frequency data](#). *Journal of the American Statistical Association*, 100(472), 1394–1411.